

Nima Soltani

Computer Engineering Student

Summary

Computer Engineering undergraduate at Shahid Beheshti University, conducting research at the Institute for Research in Fundamental Sciences (IPM) and Shahid Beheshti University. Active work in deep reinforcement learning, multimodal AI, and computational neuroscience, spanning multimodal trading agents, physics-informed neural networks, graph-based time-series modeling, blockchain/Ethereum systems, and MEG-based neural signal processing. Experienced Teaching Assistant at SBU and Sharif University of Technology. Seeking graduate research opportunities in AI and machine learning.

Research Interests

- **Graph & Hypergraph Neural Networks**
- **Financial Time-Series & Trading**
- **Computer Vision**
- **Multimodal Transformers**
- **Neurocomputing**
- **Computer Networks & Distributed Systems**

Education

2023–now **B.SC**, *Shahid Beheshti University*, Tehran, Iran
Computer Engineering, *GPA 3.6/4*

2020–2023 Diploma, National Organization for Development of Exceptional Talents (Sampad), Zanjan, Iran, *GPA 3.88/20*
Mathematics and Physics

Publications

- **N. Soltani**, A. Sadr, A. Pakniyat, K. Pakzad, and D. Rahmati, Automated Stock Trading Using Deep Reinforcement Learning and Fused xLSTM Networks, Under Review at Applied Soft Computing, Elsevier, Preprinted at *SSRN* .
- D. Rahmati, **N. Soltani**, and A. Sadr, Robust Multimodal Portfolio Trading Using Transformer-Based PPO: Technical-Fundamental Fusion with Conservative On-Policy Diffusion, Under Review at Expert Systems with Applications, Elsevier.
- A. Sadr, **N. Soltani**, V. Moghtadaiee, A. Pakniyat, D. Rahmati, and S. Gorgin, PG-KINN: A Physics-Informed Petrov-Galerkin Kolmogorov-Arnold Network for Solving Forward and Inverse PDEs, Preprinted at *arXiv* .

Research Experience

Research Assistant — Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

Dr. Dara Rahmati | 202۳ – Present

Developed deep reinforcement learning and multimodal AI systems for financial market modeling and automated trading, including **MTDP-PPO** (Transformer-based multimodal fusion with **diffusion policies**), **Fused xLSTM–PPO** trading agents, and **PG-KINN** (physics-informed Kolmogorov–Arnold networks for PDE solving), alongside **graph/hypergraph neural network** research on time-series and market dynamics.

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Research Assistant — Shahid Beheshti University, Tehran, Iran

Dr. Maedeh Mosharraf | 2024 – Present

Investigating **blockchain architectures**, **Ethereum** protocol design, and **smart contract** development (**Solidity**) for decentralized systems, with focus on **consensus mechanisms**, on-chain transaction modeling, and security analysis of **cryptocurrency and DeFi** infrastructures.

Research Assistant — Shahid Beheshti University, Tehran, Iran

Dr. Narjes Soltani | 2026 – Present

Building computational pipelines for **MEG signal preprocessing**, **source localization**, and **neural decoding**, applying machine learning and spectral/time-frequency analysis to study brain activity patterns related to **visual perception** and cognitive processing.

Teaching Experience

Academic Teaching Assistant

Sharif University of Technology

Logic Circuits | Dr. Shahin Hessabi	Feb 2026 – Jul2026
Logic Circuits | Dr. Laleh Arshadi	Feb 2026 – Jul2026
Scientific and Technical Presentation| Dr. Laleh Arshadi	Feb 2026 – Jul2026
Computer Architecture | Dr. Laleh Arshadi	Sep2025 – Feb 2026
Logic Circuits | Dr. Laleh Arshadi	Sep2025 – Feb 2026
Fundamentals of Programming | Mr. Mohammad Nazari	Sep2025 – Feb 2026
Computer Architecture | Dr. Laleh Arshadi	Jan2025 – Jun2025
Fundamentals of Programming | Dr. Marziyeh Sadri	Jan2025 – Jun2025
Fundamentals of Programming | Mr. Mohammad Zarei	Jan2025 – Jun2025
Fundamentals of Programming | Mr. Mohammad Nazari	Sep2024 – Jan2025

Shahid Beheshti University

Principles of Design and Analysis of Experiments | Dr. Narjes SoltaniDehaghani	Sep 2026 – Feb2027
Database | Dr. Hassan Haghighi	Sep 2026 – Feb2027
Engineering Probability and Statistics | Dr. Narjes SoltaniDehaghani	Sep 2026 – Feb2027
Blockchain Technology (Graduate Course) | Dr. Maedeh MosharrafDehkordi	Sep 2026 – Feb2027
Blockchain Technology (Graduate Course) | Dr. Maedeh MosharrafDehkordi	Feb 2026 – Jul2026
Linear Algebra (Head TA) | Dr. Farokhlagha Moazami Goodarzi	Feb 2026 – Jul2026
Fundamentals of Data Mining (Head TA) | Dr. Ahmad-Ali Abin	Feb 2026 – Jul2026
Advanced Programming | Dr. Mojtaba Vahidi Asl	Feb 2026 – Jul2026
Data Structures and Algorithms | Dr. Ahmad-Ali Abin	Sep2025 – Feb 2026
Fundamentals of Programming | Dr. Zeinab Pourbafarani	Sep2025 – Feb 2026
Fundamentals of Programming Course (Head TA) | Dr. Mahdi Mojahedi	Sep2025 – Feb 2026
Computer Architecture | Dr. Dara Rahmati	Sep2025 – Feb 2026
Formal Languages and Automata Theory | Dr. Ramak Ghavami Zadeh	Jan2025 – Jun2025
Computer Architecture | Dr. Razieh Salarifard	Jan2025 – Jun2025
Advanced Programming | Dr. Sadegh Aliakbari	Jan2025 – Jun2025
Advanced Programming | Dr. Mojtaba Vahidi Asl	Jan2025 – Jun2025
Fundamentals of Programming | Dr. Monireh Abdoos	Jan2025 – Jun2025
Applied Computer Skills | Dr. Hamed Malek	Sep2024 – Jan2025

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Technical Skills

- **Programming Languages:** Python, C, C++, C#, Java, MATLAB, R
- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers, Scikit-learn, OpenCV
- **Machine Learning & AI:** Deep Reinforcement Learning (PPO, Diffusion Policies), Transformer Architectures, LSTM/xLSTM, Graph & Hypergraph Neural Networks, Kolmogorov–Arnold Networks (KANs), Physics-Informed Neural Networks (PINNs)
- **Financial & Time-Series AI:** Algorithmic Trading, Portfolio Optimization, Multimodal Fusion, Financial NLP (FinBERT/FinRoBERTa)
- **Computational Neuroscience & Signal Processing:** MEG/EEG Signal Analysis, Neural Data Processing, Spectral & Time-Frequency Analysis
- **Blockchain & Distributed Systems:** Ethereum, Smart Contracts, Solidity, Blockchain Protocol Design
- **Scientific Computing & Tools:** NumPy, Pandas, SciPy, Matplotlib, Git, LaTeX, Linux, Weights & Biases, TensorBoard
- **Systems & Hardware:** Verilog, ModelSim, Raspberry Pi, Unity
- **Research Competencies:** Experimental Design, Benchmarking & Statistical Evaluation, Algorithm Design, Scientific Writing, Reproducible Research

Voluntarily Activities

- Presenter of *Pointers and Dynamic Memory Management* and *Recursion and Problem-Solving Workshops* during the Fundamentals of Programming course instructed by Dr. Mahdi Mojahedi (2025–2026).
- Presenter of *Object-Oriented Programming in Java*, *STL Containers and Iterators*, and *Game Development with Unity* Workshops during the Advanced Programming course instructed by Dr. Mojtaba Vahidi Asl (2025–2026).
- Substitute Teacher for *Instruction Set Architecture* and *MIPS Assembly Programming* during the Computer Architecture course instructed by Dr. Dara Rahmati (2025–2026).
- Presenter of *Graph Traversal Algorithms* and *Dynamic Programming Workshops* during the Data Structures and Algorithms course instructed by Dr. Ahmad-Ali Abin (2025–2026).
- Presenter of *Ethereum Smart Contract Development* and *Blockchain Security Workshop* during the Blockchain Technology (Graduate) course instructed by Dr. Maedeh MosharrafDehkordi (2026).
- Volunteer Mentor for first-year students in *programming fundamentals and algorithmic problem-solving* at Shahid Beheshti University (2025–2026).
- Secretariat Member, 1st International Conference on Artificial Intelligence, Tehran, Iran (February 2025)

Course Projects

Intelligent Game-Playing Agent (Artificial Intelligence Course)

Implemented an adversarial agent using Minimax and Alpha-Beta pruning with heuristic evaluation and depth-limited search; analyzed performance–optimality trade-offs through experimental evaluation.

Recommender System using SVD (Linear Algebra Course)

Implemented SVD via iterative numerical methods and applied PCA for latent factor modeling; built and evaluated a recommender system using RMSE and MAE metrics with visualization of embeddings

.

Virtual Memory Management Simulator (Operating Systems Course)

Developed a simulator to evaluate page replacement policies (LRU, Second Chance) and analyze the effects of page size and physical memory on page fault rates using multi-run statistical analysis.

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- **SBUMIPS Processor Design (Computer Architecture Course)**
Designed and simulated a MIPS-like processor using Verilog. Implemented instruction execution pipeline and validated correctness using ModelSim simulations.

- **Vampire Survivors-Style Game (Advanced Programming Course)**
Developed a 2D game in Unity with object-oriented architecture, real-time enemy spawning, collision detection, and player upgrade mechanics.

- **File Compression Tool (Fundamentals of Programming Course)**
Implemented a file compression utility with custom encoding scheme, focusing on algorithmic efficiency and memory management.

Honors & Awards

- Ranked within the Top 1% in the National University Entrance Exam (0.2%)
- Advanced Algorithmic Thinking and Data Structures Certificate, Quera
- ICDL Certification, Technical and Vocational Training Organization
- Academic Article Writing Workshop, Institute for Advanced Studies in Basic Sciences (IAWS–IUST)
- Serious Game Design with Cognitive Approach Workshop, Shahid Beheshti University
- Participant, *International Interdisciplinary Conference on Artificial Intelligence (IICAI)*: Engineering, Health, Finance and Humanities, Shahid Beheshti University / IEEE (May 31–June 1, 2026).
- Participant, *9th Conference on Advances in Enterprise Architecture of Iran*, Shahid Beheshti University, Tehran, Iran (November 10–12, 2025).
- Specialized Workshops, *22nd International ISC Conference on Information Security and Cryptology (ISCISC 2025)*: BIOS Security and Personalization; Fault Injection Attacks; Code Readout Attacks on Microcontrollers; Smart-Card Authentication in Access Control Systems; AI for Cybersecurity; Agent-Based AI for Automated Risk Management. Organized by the Iranian Society of Cryptology and the Cyberspace Research Institute, Shahid Beheshti University (October 6–8, 2025).
- Workshop, *Prompt Engineering for Large Language Models*, 1st International Conference on Artificial Intelligence (ICAI), Shahid Beheshti University, Tehran, Iran (February 24–25, 2025).
- Member, Iranian Bioinformatics Society (IBIS)

Languages

- Persian (Native)
- Turkish (Native)
- English (Professional Working Proficiency)

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